Phylogenetic Patterns of Brain Size and Language Complexity in Cetaceans

Introduction

Cetaceans are an order of aquatic mammals that include whales, dolphins and porpoises. They evolved from terrestrial artiodactyls into amphibians and eventually to aquatic mammals (Thewissen et al., 2009). Cetaceans, alongside primates, are often studied due to their intelligence and social behaviors, making them key models for understanding the evolution of complex cognitive traits. While the evolution of brain size has been extensively researched, there is a lack of studies that combine brain size data and communication complexity across a wide range of cetacean species. This gap in knowledge would be interesting to explore as it would provide insight into understanding the complexity of vocal and social aspects in cetaceans. This research can be applied to general knowledge on how brains have evolved to support advanced learning and communication. Cetaceans have the second largest brains relative to body size (first are humans). The evolution of large brains is often attributed to advanced cognitive capacities evolved to support complex behaviors, including communication and social interactions (Marino et al., 2007). Cetaceans have a wide range of vocal abilities, making them ideal to study evolutionary trajectory in vocal communication systems such as the evolution of rhythm. They exhibit behaviors such as vocal learning abilities, advanced breathing control, behavior synchronization and prolonged mother–infant bonds which hint at high language complexity (Hersh et al., 2024). The goal of this project is to explore the relationship between brain size and language complexity in cetaceans.

Evolutionary Question

How many times have large brain size and complex communication traits independently evolved in cetacean species?

Primary Hypothesis: Large brain sizes and complex communication traits have evolved independently multiple times within cetacean lineages due to evolutionary factors.

Methods

<u>Data Preparation:</u>

The sequencing data was sourced from the paper *Phylogenomic Resolution of the Cetacean Tree of Life Using Target Sequence Capture* (McGowen et al., 2020), available in the Dryad Digital Repository. Phenotypic data for brain size was obtained from the paper *Brain size evolution in whales and dolphins: new data from fossil mysticetes* (Marino et al., 2021).
To identify overlapping species, the taxa lists from both sources were compared, resulting in 36 shared species. A new PHYLIP file was created containing these species and sequences were trimmed to include seven genes of interest: AQP6, FBXO2, FGG, FOXP1, HOXA13, HOXA9, HOXB7. Gene partitions were provided from the Dryad repository. Due to lower computational power and time constraints, the dataset was further filtered to 19 species with the highest sequence quality(lowest ambiguities) and *Hippopotamus amphibius* was included as an outgroup. The hippopotamus was included as the outgroup because sequence data was available in the same source as the other data, and it shares evolutionary history with cetaceans. The PHYLIP file was converted into FASTA format for analysis. Brain mass data for the hippopotamus was sourced from the paper *Delphinid brain development from neonate to adulthood with comparisons to other cetaceans and artiodactyls* (Ridgway et al., 2018).

Phylogenetic Tree Construction with RAxML:

To construct a phylogenetic tree, RAxML was selected. The FASTA file was aligned using Clustal Omega, chosen due to its ability to handle sequences with gaps and its reliability over older tools like ClustalW. After alignment, the file was converted back to PHYLIP format for RAxML processing. Gene partitions from the Dryad repository were adjusted to account for sequence length changes post-alignment. A model test was run on the sequences and determined that GTR+G(4)+I had the lowest BIC score and a lowest AIC score, therefore that was the model used in our RAxML. GTR allows for variable substitution rates between nucleotides, G(4) accounts for gamma-distributed rate variation across sites, and I proportions of invariant sites. Other notable parameters used were 1000 bootstrap replications, 9503 seeds for bootstrap and 9004 for tree searching algorithms, and the hippopotamus as the outgroup. The tree was then plotted with bootstrap values.

Continuous Maximum Likelihood Model:

Using the RAxML tree, brain mass data (log-transformed for proportional scaling) was mapped onto the phylogeny. A heatmap was generated to visualize the relationships between species and brain size.

Subset Data Preparation for Language Complexity:

A subset of eight cetacean species was selected from the 19 species dataset, based on available data for vocal and social complexity. The *Hippopotamus amphibius* remained as the outgroup. Two separate scales for vocalization and social complexity were developed:

Vocalization Complexity:

1 - Minimal vocalizations or no clear evidence of diverse sound production.

2 - Simple vocal repertoire with limited sound types.

3 - Moderate diversity of sound types, limited use in communication.

4 - Diverse repertoire used for communication or coordination.

5 - Highly complex repertoire, with evidence of vocal learning, distinct dialects, or sophisticated use.

Social Complexity:

1 - Solitary or minimally social; no clear cooperative behaviors.

2 - Small groups with limited cooperation or social structure.

3 - Moderate group size (10–30 individuals) with some cooperation.

4 - Large groups (30–100+ individuals) with evidence of specialization or cooperative behaviors.

5 - Highly social, showing evidence of culture, cooperation, or long-term bonds.

The scores from both scales were combined to generate a language complexity score for each species.

Subset Phylogenetic Analysis:

The same RAxML pipeline was used to construct a phylogenetic tree for the subset dataset, incorporating updated partitions.

Discrete Ancestral State Reconstruction:

Ancestral state reconstruction was performed on the RAxML tree to explore the evolution of language complexity and brain size as traits. A custom transition matrix was created to represent the evolutionary costs of state changes. The analysis was repeated for both traits to infer ancestral states and evolutionary pathways.

Results

The species identification table found below as Table 1 can be used to assist in understanding the results in this study.

Table 1: Species Identification Table

Species Name	Identifier	Common Name
Balaenoptera borealis	Bbor	Sei Whale
Balaenoptera musculus	Bmus49099	Blue Whale
Balaenoptera physalus	Bphy_baits	Fin Whale
Caperea marginata	Cmar59905	Pygmy Right Whale
Delphinus delphis	DdelSJR	Common Dolphin
Globicephala macrorhynchus	Gmac39091	Long-finned Pilot Whale
Grampus griseus	Ggri	Risso's Dolphin
Kogia sima	Ksim	Dwarf Sperm Whale
Lagenorhynchus albirostris	Lalb	White-beaked Dolphin
Lagenorhynchus obscurus	Lobs	Dusky Dolphin
Lipotes vexillifer	Lvex	Yangtze River Dolphin (Baiji)
Mesoplodon europaeus	Meur	Gervais' Beaked Whale
Orcinus orca	Oorc	Killer Whale (Orca)
Physeter macrocephalus	Pmac	Sperm Whale
Pseudorca crassidens	Pcra123188	False Killer Whale
Stenella clymene	Scly1726	Clymene Dolphin
Stenella coeruleoalba	Scoe	Striped Dolphin
Steno bredanensis	Sbre116871	Rough-toothed Dolphin
Tursiops truncatus	Ttru	Bottlenose Dolphin
Hippopotamus amphibius	Hippo	Hippopotamus

The RAxML demonstrated in Figure 1 shows the bootstrap values for all 20 species. The highest bootstrap value is 1 and the lowest is 0.111, with 83% of values being higher than 0.50.

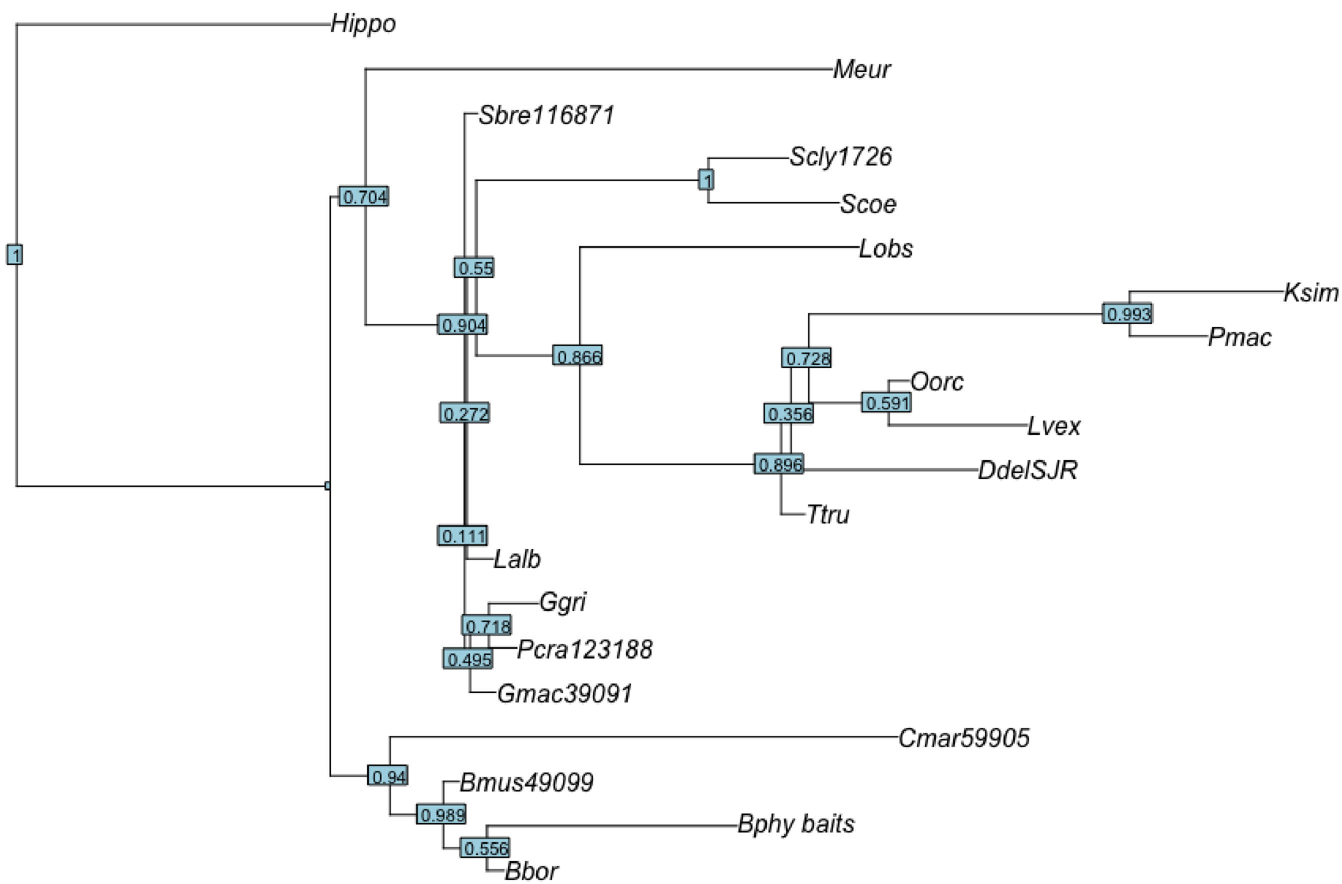

Figure 1:RAxML tree of concatenated sequences for 20 cetacean species with Hippo as the outgroup. Bootstrap values are indicated at branch nodes.

The continuous ancestral state reconstruction (ASR) for brain size is visualized in Figure 2. Brain size among cetaceans ranges from 2.76 to 3.89 (log-transformed values), with a branch length of 0.052.

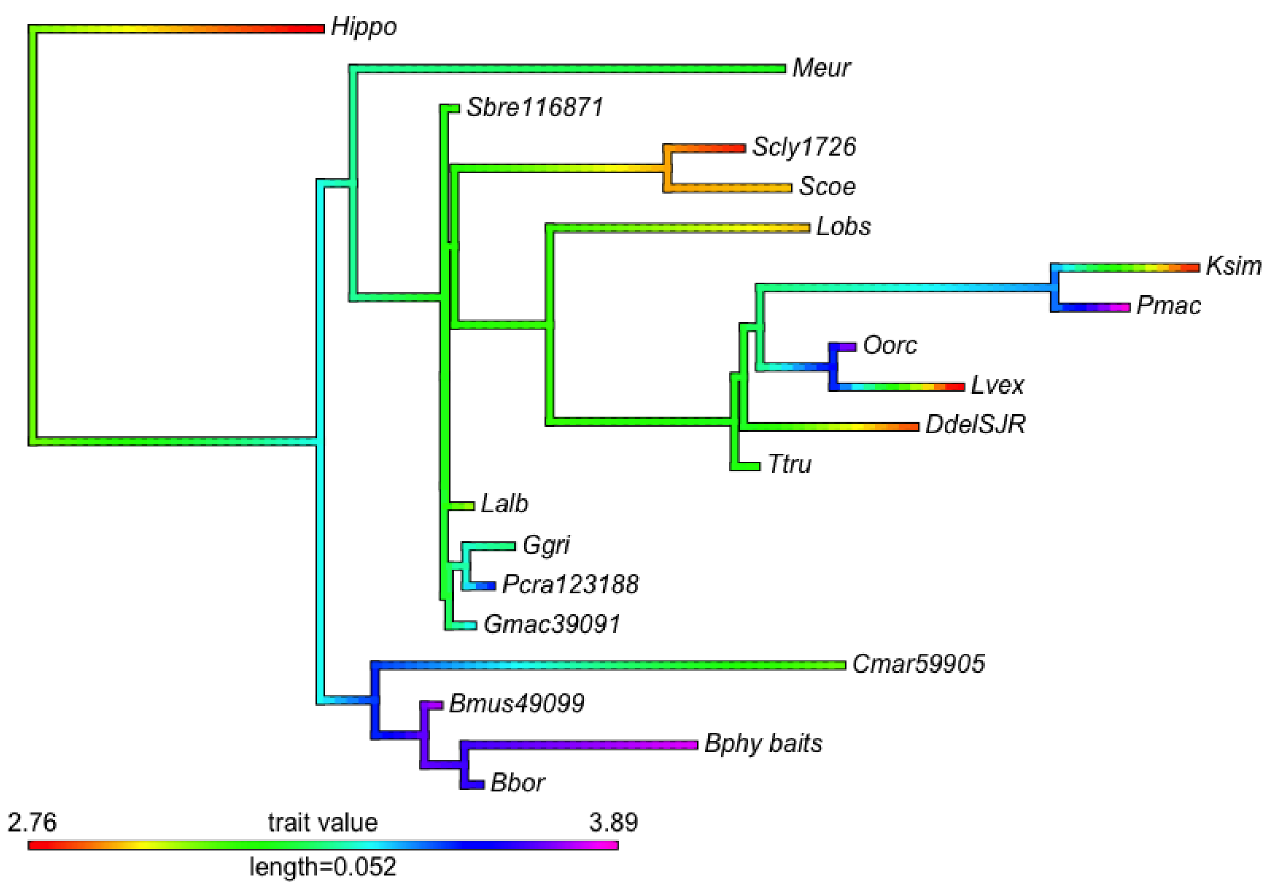

Figure 2: Heatmap representing continuous ancestral state reconstruction of brain size (log-transformed) for cetacean species. Cool toned colors like purple and blue represent large brain size.

The RAxML demonstrated in Figure 3 shows the bootstrap values for the subset of species chosen for further analysis. The highest bootstrap value is 1 and the lowest is 0.453. All except the lowest value are above 0.92.

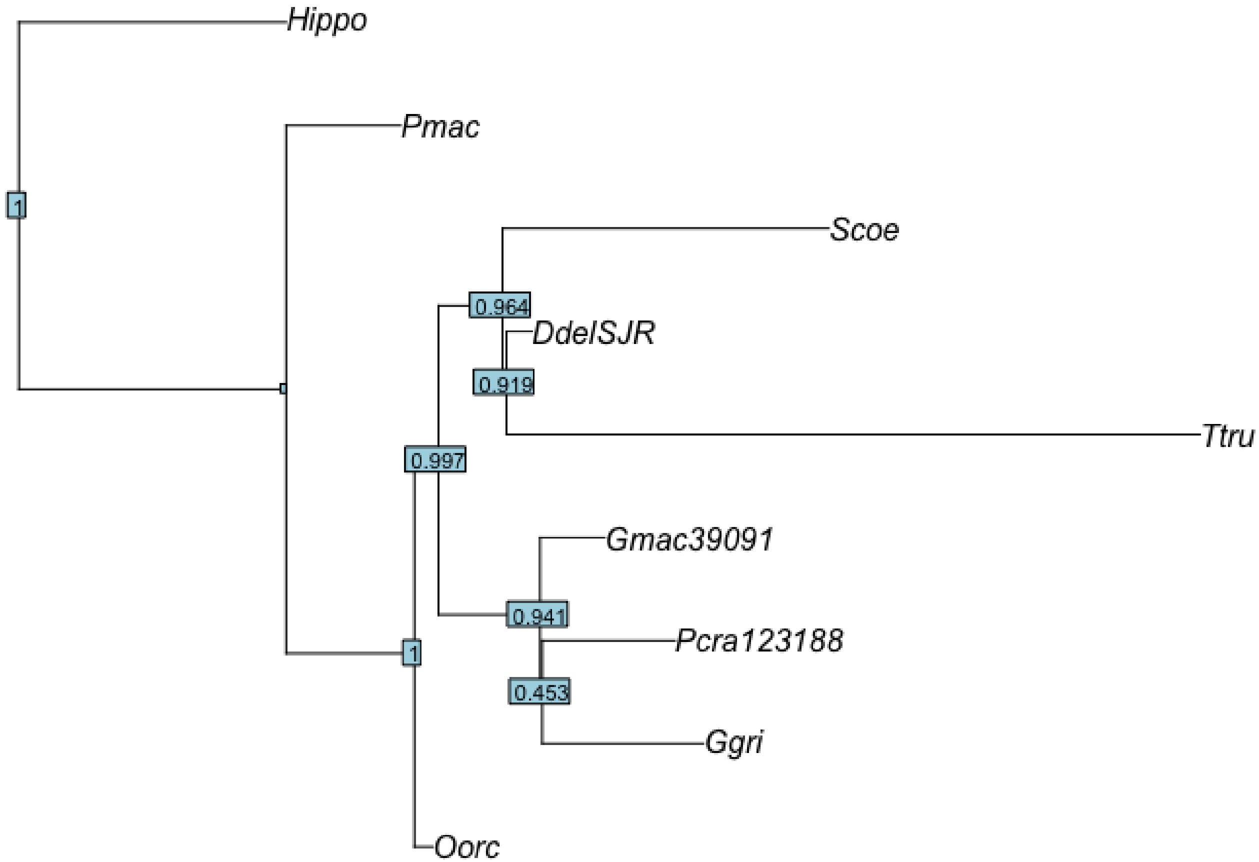

Figure 3: RAxML tree of concatenated sequences for subset species, including Hippo as the outgroup. Bootstrap values are indicated at branch nodes.

Ancestral state reconstruction for brain size (Figure 4) shows four distinct states. 77% of species exhibit a brain size larger than 3.08 (log-transformed).

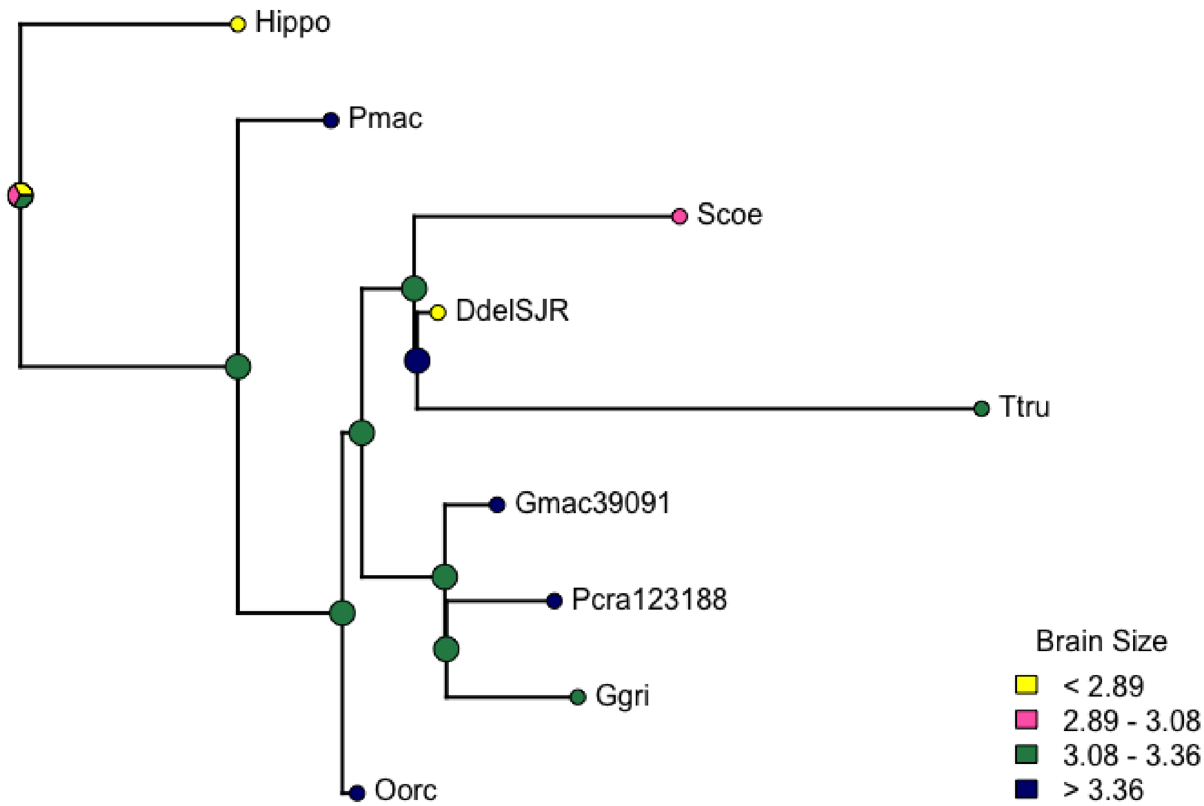

Figure 4: Ancestral state reconstruction of brain size for subset species

Ancestral state reconstruction for language complexity (Figure 5) reveals five distinct states, all of which appear in the phylogenetic tree. The majority of species (77%) exhibit a language complexity level of 3 or higher.

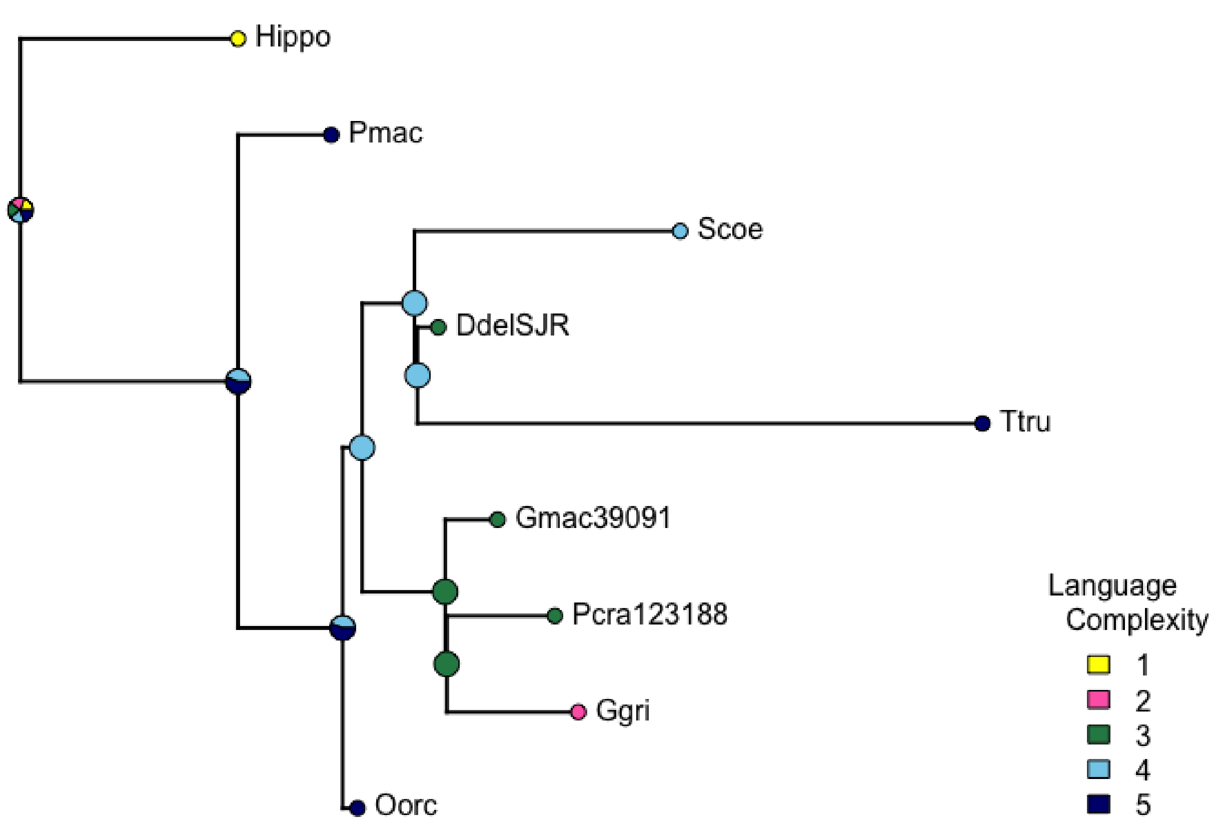

Figure 5:Ancestral state reconstruction of language complexity for subset species.

Discussion

With 83% of the values higher than 0.50 in Figure 1, we can be relatively confident that the relationships between species are correct. The species with lower bootstrap support are Lalb and DdelSJR. Based on the table below, we can conclude that the relative relationships between the White-Beaked Dolphin (Lalb) and the Common Dolphin (DdelSJR) are not as certain. The clades depicted are logical, as species are grouped with their respective families. For example, the Balaenidae family includes the Blue Whale (Bmus49099), Sei Whale (Bbor), Fin Whale (Bphy_baits), and Pygmy Right Whale (Cmar59905). The Gervais' Beaked Whale (Meur) belongs to the Ziphiidae family, while the rest of the species are in a Delphinidae clade.

The ASR of brain sizes utilized a heatmap, with warm colors such as red and orange representing smaller brain sizes and cool colors like purple and dark blue representing larger brain sizes. Looking at the cool colors, we can see that larger brain size appears to have evolved independently multiple times:

1. In the Balaenidae family clade (Bmus49099, Bbor, Bphy_baits, and Cmar59905),
2. In the Ggri, Pcra123188, and Gmac39091 clade,
3. Independently in the Orca (Oorca),
4. And one last time in the Sperm Whale (Pmac).

Therefore large brain size evolved independently four times. This independent evolution is supported since lineages with larger brains are not closely related, except for the Oorca and Pmac clade. Interestingly, large brain size was not conserved within that clade.

High bootstrap values in Figure 3 provide strong support for the maximum likelihood tree created, although there is some uncertainty regarding the placement of Risso's Dolphin (Ggri) with a bootstrap value of 0.453. Figure 4 supports our findings in Figure 2, showing four distinct species in which large brain size independently evolved (Oorca, Pmac, Gmac39091, and Pcra123188). However because this is a small subset of species and the ASR graph is discrete rather than continuous, the findings are less impactful. Figure 4 was primarily created to provide a visual side-by-side comparison with Figure 5 to help determine if there is overlap between lineages with large brain sizes and those with high language complexity.

In Figure 5, higher language complexity is shown in navy blue. High levels of language complexity likely evolved two times:

1. In the Sperm Whale (Pmac) and the Orca (Oorca),
2. And in the Bottlenose Dolphin (Ttru).

There is a potential that higher language complexity evolved three separate times, as the pie matrix shows a 44% chance that language complexity decreased from level 5 to level 4 between the Sperm Whale and Orca lineage. However, it is 56% more likely that the trait of high language complexity was conserved. The data may be skewed due to an information bias—species with higher language complexity are more likely to be studied, leading to more data being available and increasing the chances of being chosen for the subset.

When comparing the evolution of brain size and language complexity, there is a link supporting co-evolution; however, this is not always the case. A notable difference is seen in the Bottlenose Dolphin (Ttru), which has a brain size in the upper medium range but demonstrates high language complexity. The opposite is true for the clade containing Gmac39091, Pcra123188, and Ggri, which had a large comparative brain size but exhibited average or lower language complexity. This indicates that while large brain sizes and language complexity may be correlated in some cases, they also appear to evolve independently in others.

Conclusion

From our phylogenetic analysis, we can conclude that brain size likely evolved independently four times in cetacean species, and complex communication likely evolved independently two times. The hypothesis that large brain sizes and complex communication traits have evolved independently multiple times within cetacean lineages due to evolutionary factors is supported by the findings of this study. For example, despite Bottlenose Dolphins (Ttru) not having as large a brain size compared to other members of the Delphinidae family, they might have developed high levels of language complexity due to evolutionary pressures. Bottlenose dolphins inhabit a wide variety of habitats, ranging from harbors and coastal waters to far offshore in the open ocean (National Oceanic and Atmospheric Administration, n.d.). They are one of the most diverse dolphin species, which could explain why they may have developed higher language complexity as an adaptation to environmental pressures.

Exploring the evolution of brain size and complex communication in cetaceans provides a deeper understanding of their phylogenetic history. More interestingly, it can offer insight into how culture and language learning evolved in certain cetaceans, such as Orcas and Sperm Whales. Language Artificial Intelligence, like the technology that powers Google Translate, has the potential to be applied to decipher animal languages—essentially opening communication with animals. Project CETI (Cetacean Translation Initiative) is an animal-translation project focused on a clan of sperm whales, where acoustics are collected to assemble a database that AI can then use to interpret language rules and apply those patterns to translate sentences it has not previously encountered (Project CETI, n.d.).

One limitation of this study is its scope; due to time and computational resource constraints, only a limited number of species and their genes were analyzed. Expanding the project to include all 36 cetacean species and their full genomes would be valuable to assess whether different results emerge.Another study suggested that during the Archaeoceti-to-modern cetacean faunal transition, water temperature exerted selective pressure on cetacean brain evolution (Manger PR 2006). Conducting a coalescent time analysis could help determine if our findings align with this study and identify other evolutionary factors that may have contributed to the evolution of large brain size and complex language. Performing divergence time estimation would further allow us to examine whether the evolution of these traits coincides with environmental or ecological changes.

Supplementary Material

R code and graphs provided in a PDF of RMD: final.pdf

Language Complexity Scale in a PDF: Cetaceans Language complexity.pdf

Works Cited

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